



MEMPHIS, TN

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Problem: Flood in Memphis, Tennessee

- Memphis sits on the Mississippi River floodplain, low flat land where three major waterways connect.
- When intense rain happens, river levels can rise and push water into neighborhoods, roads, and businesses.
- For instance, in 2011, the Mississippi River almost reached its highest level recorded in Memphis, risking thousands of people and buildings.
- Flooding can damage homes, cars, schools, and roads. It can contaminate water and costs millions to repair.
- Memphis has a lot of pavement and buildings instead of natural ground for water to soak into, increasing the severity of urban flooding.

Solution: How Memphis Reduces Flood Risk

- Large levees and concrete floodwalls line parts of the Mississippi River and local streams to prevent flooding around homes and businesses.
- Memphis has many flood pumping stations that move rainwater over the levees into the river.
- Memphis utilizes a separate storm sewer system that carries rainwater to nearby creeks and rivers to reduce street flooding.
- Wetlands and floodplains are being restored along the Wold River and its greenway to provide more space for floodwater to safely spread out.
- Local building codes restrict where and how people can build, requiring higher floors or stronger foundations in flood-prone areas.
- Agencies watch river forecasts and issue warnings and evacuations before major flooding.

Effect: Result of the Flood Plan

- Levees, floodwalls, and pumping stations proved very successful during the Mississippi River flood of 2011, where most of Memphis stayed protected.
- Fewer homes and businesses are severely damaged during major floods, which saves money on repairs.
- Restored wetlands and green spaces absorb stormwater, improve water quality, and provide wildlife habitat for residents.
- Flood maps, zoning, and building standards help people avoid building in dangerous areas.
- Overall, the Memphis flood plan helps the city stay safer and better prepared for future flooding as weather patterns continue to change.